



Machine Learning



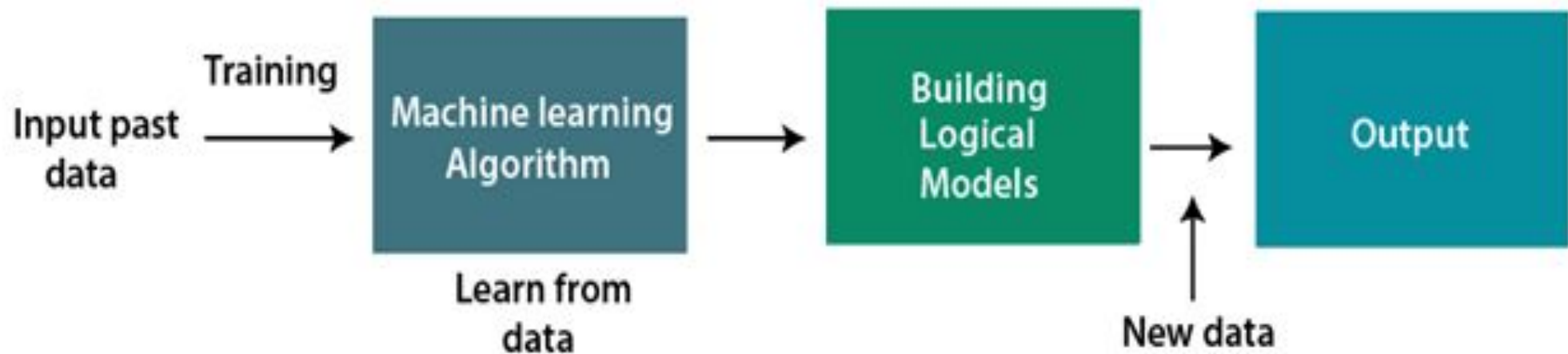
Module 1

Introduction to ML

Machine Learning

- Machine learning (ML) is a branch of artificial intelligence (AI) that enables computers to “self-learn” from training data and improve over time, without being explicitly programmed.
- Machine learning algorithms are able to detect patterns in data and learn from them, in order to make their own predictions.
- In short, machine learning algorithms and models learn through experience.

How does Machine Learning work?





Features of Machine Learning:

- Machine learning uses data to detect various patterns in a given dataset.
- It can learn from past data and improve automatically.
- It is a data-driven technology.
- Machine learning is much similar to data mining as it also deals with the huge amount of the data.



Need for Machine Learning:

- Rapid increment in the production of data
- Solving complex problems, which are difficult for a human
- Decision making in various sector including finance
- Finding hidden patterns and extracting useful information from data.

Applications of Machine Learning

1. Image Recognition:

- used to identify objects, persons, places, digital images, etc.
- The popular use case of image recognition and face detection is, **Automatic friend tagging suggestion**
- Facebook provides us a feature of auto friend tagging suggestion.
- the technology behind this is machine learning's **face detection and recognition algorithm**.

Applications of Machine Learning

2. Speech Recognition

- In Google: **"Search by voice"**
- Speech recognition is a process of converting voice instructions into text, and it is also known as **"Speech to text"**, or **"Computer speech recognition."**
- **Google assistant, Siri, Cortana, and Alexa**

Applications of Machine Learning

3. Traffic prediction

- Google Maps: shows the correct path with the shortest route and predicts the traffic conditions.
- It predicts the traffic conditions such as whether traffic is cleared, slow-moving, or heavily congested with the help of two ways:
 - 1. Real Time location** of the vehicle from Google Map app and sensors
 - 2. Average time has taken** on past days at the same time.

Applications of Machine Learning

4. Product recommendations

- widely used by various e-commerce and entertainment companies such as **Amazon**, **Netflix**, etc., for product recommendation to the user
- searching for some product on Amazon->started getting an advertisement for the same product while internet surfing on the same browser
- Netflix, some recommendations for entertainment series, movies, etc.,

Applications of Machine Learning

5. Self-driving cars

- Tesla, the most popular car manufacturing company is working on self-driving car.
- It is using unsupervised learning method to train the car models to detect people and objects while driving.

6. Email Spam and Malware Filtering

- new email is filtered automatically as important, normal, and spam.
- We always receive an important mail in our inbox and spam emails in our spam box

Applications of Machine Learning

7. Virtual Personal Assistant

- various virtual personal assistants such as **Google assistant, Alexa, Cortana, Siri.**
- helps in finding the information using voice instruction
- helps in various ways just by voice instructions such as Play music, call someone, Open an email, Scheduling an appointment, etc.
- These assistant record our voice instructions, send it over the server on a cloud, and decode it using ML algorithms and act accordingly.

Applications of Machine Learning

8. Online Fraud Detection

- Machine learning is making our online transaction safe and secure by detecting fraud transaction.
- Whenever we perform some online transaction, there may be various ways that a fraudulent transaction can take place.
- such as **fake accounts**, **fake ids**, and **steal money** in the middle of a transaction.
- So to detect this, **Feed Forward Neural network** helps us by checking whether it is a genuine transaction or a fraud transaction.

Applications of Machine Learning

9. Stock Market trading

- Machine learning is widely used in stock market trading.
- In the stock market, there is always a risk of up and downs in shares, so for this machine learning's long short term memory neural network is used for the prediction of stock market trends.

10. Medical Diagnosis:

- In medical science, machine learning is used for diseases diagnoses. With this, medical technology is growing very fast and able to build 3D models that can predict the exact position of lesions in the brain.
- It helps in finding brain tumors and other brain-related diseases easily.

Applications of Machine Learning

11. Automatic Language Translation

- converting the text into our known languages
- Google's GNMT (Google Neural Machine Translation) , a Neural Machine Learning that translates the text into our familiar language, and it called as automatic translation.

Common issues in Machine Learning

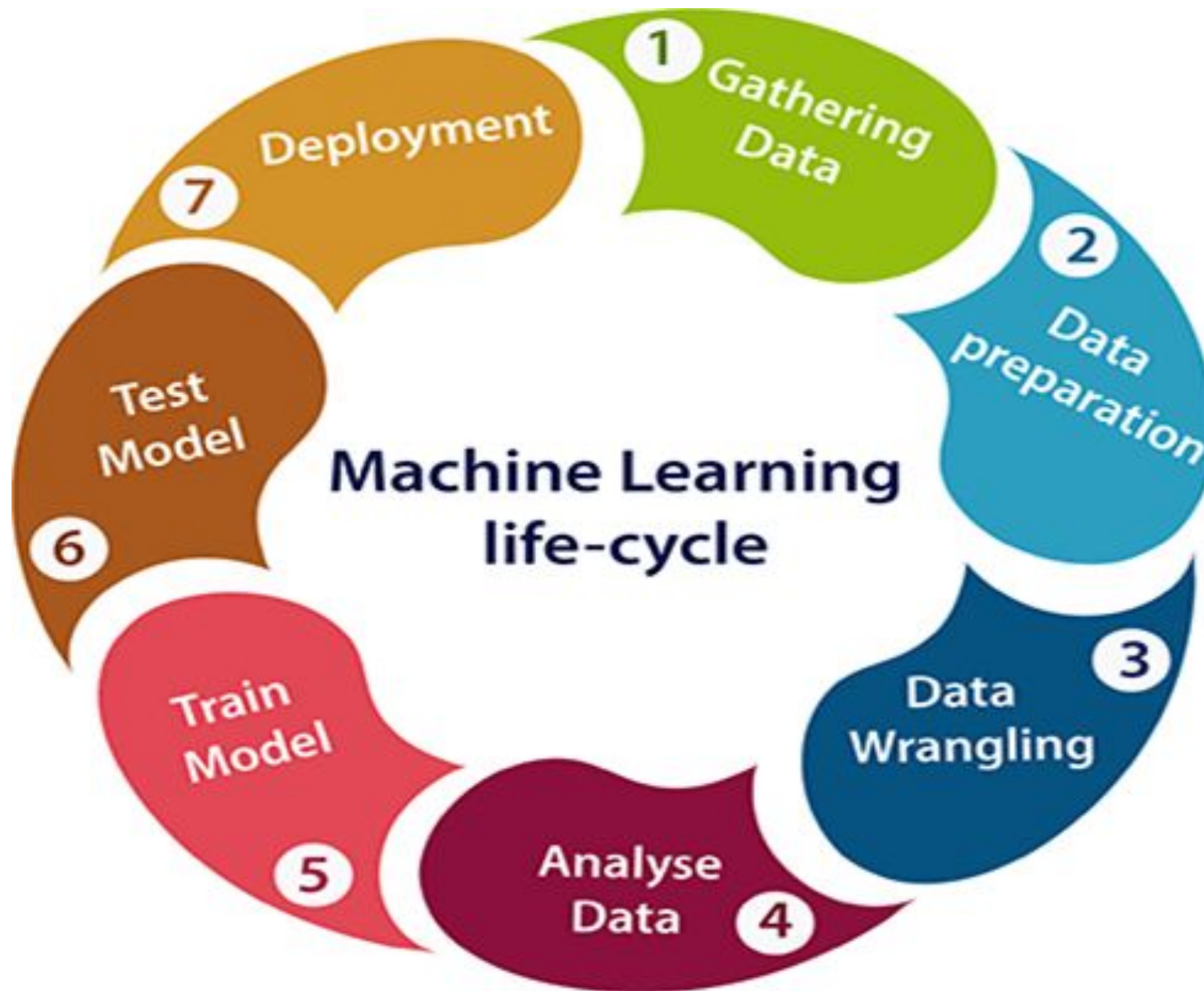
- 1. Inadequate Training Data**
- 2. Poor quality of data**
- 3. Non-representative training data**
- 4. Overfitting and Underfitting**
- 5. Monitoring and maintenance**
- 6. Getting bad recommendations**
- 7. Lack of skilled resources**
- 8. Process Complexity of Machine Learning**
- 9. Slow implementations and results**
- 10. Irrelevant features**



Machine learning Life cycle

- **Gathering Data**
- **Data preparation**
- **Data Wrangling**
- **Analyse Data**
- **Train the model**
- **Test the model**
- **Deployment**

Machine learning Life cycle



Machine learning Life cycle

1. Gathering Data:

- to identify and obtain all data-related problems.
- identify the different data sources, as data can be collected from various sources such as **files**, **database**, **internet**, or **mobile devices**.
- The quantity and quality of the collected data will determine the efficiency of the output. The more will be the data, the more accurate will be the prediction.
- This step includes:
 - **Identify various data sources**
 - **Collect data**
 - **Integrate the data obtained from different sources**
- By performing the above task, we get a coherent set of data, also called as a **dataset**. It will be used in further steps.

Machine learning Life cycle

2. Data preparation

- After collecting the data, prepare it for further steps. Data preparation is a step where we put our data into a suitable place and prepare it to use in our machine learning training.
- first, put all data together, and then randomize the ordering of data.
- This step can be further divided into two processes:

- **Data exploration:**

It is used to understand the nature of data that we have to work with. We need to understand the characteristics, format, and quality of data. A better understanding of data leads to an effective outcome. In this, we find Correlations, general trends, and outliers.

- **Data pre-processing:**

Now the next step is preprocessing of data for its analysis.

Machine learning Life cycle

3. Data Wrangling

- the process of cleaning and converting raw data into a useable format. It is the process of cleaning the data, selecting the variable to use, and transforming the data in a proper format to make it more suitable for analysis in the next step.
- It is not necessary that data we have collected is always of our use as some of the data may not be useful.
- In real-world applications, collected data may have various issues, including: **Missing Values, Duplicate data, Invalid data, Noise**
- So, we use various filtering techniques to clean the data.
- It is mandatory to detect and remove the above issues because it can negatively affect the quality of the outcome.

Machine learning Life cycle

4. Data Analysis

- Now the cleaned and prepared data is passed on to the analysis step. This step involves:
 - **Selection of analytical techniques**
 - **Building models**
 - **Review the result**
- The aim of this step is to build a machine learning model to analyze the data using various analytical techniques and review the outcome.
- It starts with the determination of the type of the problems, where we select the machine learning techniques such as **Classification, Regression, Cluster analysis, Association**, etc.
- Then build the model using prepared data, and evaluate the model.
- Hence, in this step, we take the data and use machine learning algorithms to build the model.

Machine learning Life cycle

5. Train Model

- train our model to improve its performance for better outcome of the problem.
- use datasets to train the model using various machine learning algorithms. Training a model is required so that it can understand the various patterns, rules, and, features.

6. Test Model

- Once our machine learning model has been trained on a given dataset, then we test the model. In this step, we check for the accuracy of our model by providing a test dataset to it.
- Testing the model determines the percentage accuracy of the model as per the requirement of project or problem.

Machine learning Life cycle

7. Deployment

- deploy the model in the real-world system.
- If the above-prepared model is producing an accurate result as per our requirement with acceptable speed, then we deploy the model in the real system.
- But before deploying the project, we will check whether it is improving its performance using available data or not. The deployment phase is similar to making the final report for a project.



Types of Machine Learning

1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Semi-Supervised Machine Learning
4. Reinforcement Learning

1. *Supervised Machine Learning*

- train the machines using the "labelled" dataset, and based on the training, the machine predicts the output.
- first, train the machine with the input and corresponding output, and then ask the machine to predict the output using the test dataset.
- **The main goal of the supervised learning technique is to map the input variable(x) with the output variable(y).**
- Some real-world applications of supervised learning are **Risk Assessment, Fraud Detection, Spam filtering, etc.**

Categories of Supervised Machine Learning

- Supervised machine learning can be classified into two types of problems:

- a. **Classification**

- b. **Regression**

a) **Classification**

- Classification algorithms are used to solve the classification problems in which the output variable is categorical, such as **"Yes" or No, Male or Female, Red or Blue, etc.**
- predict the categories present in the dataset.
- real-world examples of classification algorithms are **Spam Detection, Email filtering, etc.**

Categories of Supervised Machine Learning

b) Regression

- Regression algorithms are used to solve regression problems in which there is a linear relationship between input and output variables.
- These are used to predict continuous output variables, such as market trends, weather prediction, etc.

Advantages and Disadvantages of Supervised Learning

■ **Advantages:**

- Since supervised learning work with the labelled dataset so we can have an exact idea about the classes of objects.
- These algorithms are helpful in predicting the output on the basis of prior experience.

■ **Disadvantages:**

- These algorithms are not able to solve complex tasks.
- It may predict the wrong output if the test data is different from the training data.
- It requires lots of computational time to train the algorithm.

Applications of Supervised Learning

- **Image Segmentation:**

Supervised Learning algorithms are used in image segmentation. In this process, image classification is performed on different image data with pre-defined labels.

- **Medical Diagnosis:**

Supervised algorithms are also used in the medical field for diagnosis purposes. It is done by using medical images and past labelled data with labels for disease conditions. With such a process, the machine can identify a disease for the new patients.

- **Fraud Detection** - Supervised Learning classification algorithms are used for identifying fraud transactions, fraud customers, etc. It is done by using historic data to identify the patterns that can lead to possible fraud.

- **Spam detection** - In spam detection & filtering, classification algorithms are used. These algorithms classify an email as spam or not spam. The spam emails are sent to the spam folder.

- **Speech Recognition** - Supervised learning algorithms are also used in speech recognition. The algorithm is trained with voice data, and various identifications can be done using the same, such as voice-activated passwords, voice commands, etc.

1. *Unsupervised Machine Learning*

- the machine is trained using the unlabeled dataset, and the machine predicts the output without any supervision.
- the models are trained with the data that is neither classified nor labelled, and the model acts on that data without any supervision.
- **The main aim of the unsupervised learning algorithm is to group or categories the unsorted dataset according to the similarities, patterns, and differences.** Machines are instructed to find the hidden patterns from the input dataset.

Categories of Unsupervised Machine Learning

- Unsupervised Learning can be further classified into two types, which are given below:
 - a. **Clustering**
 - b. **Association**

a) Clustering

- The clustering technique is used when we want to find the inherent groups from the data.
- It is a way to group the objects into a cluster such that the objects with the most similarities remain in one group and have fewer or no similarities with the objects of other groups.
- An example of the clustering algorithm is grouping the customers by their purchasing behaviour.

Categories of Unsupervised Machine Learning

b) Association

- Association rule learning is an unsupervised learning technique, which finds interesting relations among variables within a large dataset.
- The main aim of this learning algorithm is to find the dependency of one data item on another data item and map those variables accordingly so that it can generate maximum profit.
- This algorithm is mainly applied in **Market Basket analysis, Web usage mining, continuous production, etc.**

Advantages and Disadvantages of Unsupervised Learning Algorithm

■ **Advantages:**

- can be used for complicated tasks compared to the supervised ones because these algorithms work on the unlabeled dataset.
- Unsupervised algorithms are preferable for various tasks as getting the unlabeled dataset is easier as compared to the labelled dataset.

■ **Disadvantages:**

- The output of an unsupervised algorithm can be less accurate as the dataset is not labelled, and algorithms are not trained with the exact output in prior.
- Working with Unsupervised learning is more difficult as it works with the unlabelled dataset that does not map with the output.

Applications of Unsupervised Learning

- **Network Analysis:** Unsupervised learning is used for identifying plagiarism and copyright in document network analysis of text data for scholarly articles.
- **Recommendation Systems:** Recommendation systems widely use unsupervised learning techniques for building recommendation applications for different web applications and e-commerce websites.
- **Anomaly Detection:** Anomaly detection is a popular application of unsupervised learning, which can identify unusual data points within the dataset. It is used to discover fraudulent transactions.
- **Singular Value Decomposition:** Singular Value Decomposition or SVD is used to extract particular information from the database. For example, extracting information of each user located at a particular location.

3. *Semi-Supervised Learning*

- **Semi-Supervised learning is a type of Machine Learning algorithm that lies between Supervised and Unsupervised machine learning.**
- It represents the intermediate ground between Supervised (With Labelled training data) and Unsupervised learning (with no labelled training data) algorithms and uses the combination of labelled and unlabeled datasets during the training period.
- The main aim of semi-supervised learning is to effectively use all the available data, rather than only labelled data like in supervised learning.
- Initially, similar data is clustered along with an unsupervised learning algorithm, and further, it helps to label the unlabeled data into labelled data.

Advantages and disadvantages of Semi-supervised Learning

■ **Advantages:**

- It is simple and easy to understand the algorithm.
- It is highly efficient.
- It is used to solve drawbacks of Supervised and Unsupervised Learning algorithms.

■ **Disadvantages:**

- We cannot apply these algorithms to network-level data.
- Accuracy is low.

4. Reinforcement Learning

- Reinforcement learning works on a feedback-based process, in which an AI agent (A software component) automatically explore its surrounding by hitting & trail, taking action, learning from experiences, and improving its performance.
- Agent gets rewarded for each good action and get punished for each bad action; hence the goal of reinforcement learning agent is to maximize the rewards.
- In reinforcement learning, there is no labelled data like supervised learning, and agents learn from their experiences only.

4. Reinforcement Learning

- The reinforcement learning process is similar to a human being; for example, a child learns various things by experiences in his day-to-day life.
- An example of reinforcement learning is to play a game, where the Game is the environment, moves of an agent at each step define states, and the goal of the agent is to get a high score. Agent receives feedback in terms of punishment and rewards.
- Due to its way of working, reinforcement learning is employed in different fields such as **Game theory, Operation Research, Information theory, multi-agent systems.**
- A reinforcement learning problem can be formalized using **Markov Decision Process(MDP)**. In MDP, the agent constantly interacts with the environment and performs actions; at each action, the environment responds and generates a new state.

Real-world Use cases of Reinforcement Learning

- **Video Games:**

RL algorithms are much popular in gaming applications. It is used to gain super-human performance.

- **Resource Management:**

to automatically learn and schedule resources to wait for different jobs in order to minimize average job slowdown.

- **Robotics:**

Robots are used in the industrial and manufacturing area, and these robots are made more powerful with reinforcement learning. There are different industries that have their vision of building intelligent robots using AI and Machine learning technology.

- **Text Mining:**

Text-mining, one of the great applications of NLP, is now being implemented with the help of Reinforcement Learning by Salesforce company.

Advantages and Disadvantages of Reinforcement Learning

■ Advantages

- It helps in solving complex real-world problems which are difficult to be solved by general techniques.
- The learning model of RL is similar to the learning of human beings; hence most accurate results can be found.
- Helps in achieving long term results.

■ Disadvantages

- RL algorithms are not preferred for simple problems.
- RL algorithms require huge data and computations.
- Too much reinforcement learning can lead to an overload of states which can weaken the results.

Performance Metrics in Machine Learning

- ***To evaluate the performance or quality of the model, different metrics are used, and these metrics are known as performance metrics or evaluation metrics.***
- These performance metrics help us understand how well our model has performed for the given data.
- In this way, we can improve the model's performance by tuning the hyper(top level)-parameters.
- Each ML model aims to generalize well on unseen/new data, and performance metrics help determine how well the model generalizes on the new dataset.

1. Performance Metrics for Classification

- In a classification problem, the category or classes of data is identified based on training data.
- The model learns from the given dataset and then classifies the new data into classes or groups based on the training.
- It predicts class labels as the output, such as *Yes or No*, *0 or 1*, *Spam or Not Spam*, etc.
- To evaluate the performance of a classification model, different metrics are used, and some of them are as follows:
 - **Accuracy**
 - **Confusion Matrix**
 - **Precision**
 - **Recall**
 - **F-Score**

1. Performance Metrics for Classification

■ I. Accuracy

The accuracy metric is one of the simplest Classification metrics to implement, and it can be determined as the number of correct predictions to the total number of predictions.

- It can be formulated as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total number of predictions}}$$

■ **When to Use Accuracy?**

- It is good to use the Accuracy metric when the target variable classes in data are approximately balanced.
- For example, if 60% of classes in a fruit image dataset are of Apple, 40% are Mango. In this case, if the model is asked to predict whether the image is of Apple or Mango, it will give a prediction with 97% of accuracy.

■ **When not to use Accuracy?**

- It is recommended not to use the Accuracy measure when the target variable majorly belongs to one class.
- For example, Suppose there is a model for a disease prediction in which, out of 100 people, only five people have a disease, and 95 people don't have one.
- In this case, if our model predicts every person with no disease (which means a bad prediction), the Accuracy measure will be 95%, which is not correct.

II. Confusion Matrix

- A confusion matrix is a tabular representation of prediction outcomes of any binary classifier, which is used to describe the performance of the classification model on a set of test data when true values are known.
- A typical confusion matrix for a binary

n=165	Predicted: NO	Predicted: YES
	50	10
Actual: NO	5	100

- We can determine the following from the above matrix:
 - In the matrix, columns are for the prediction values, and rows specify the Actual values. Here Actual and prediction give two possible classes, Yes or No. So, if we are predicting the presence of a disease in a patient, the Prediction column with Yes means, Patient has the disease, and for NO, the Patient doesn't have the disease.
 - In this example, the total number of predictions are 165, out of which 110 time predicted yes, whereas 55 times predicted No.
 - However, in reality, 60 cases in which patients don't have the disease, whereas 105 cases in which patients have the disease.

- 
- In general, the table is divided into four terminologies:
 1. **True Positive(TP):** In this case, the prediction outcome is true, and it is true in reality, also.
 2. **True Negative(TN):** in this case, the prediction outcome is false, and it is false in reality, also.
 3. **False Positive(FP):** In this case, prediction outcomes are true, but they are false in actuality.
 4. **False Negative(FN):** In this case, predictions are false, and they are true in actuality.

■ III. Precision

- The precision metric is used to overcome the limitation of Accuracy.
- The precision determines the proportion of positive prediction that was actually correct.
- It can be calculated as the True Positive or predictions that are actually true to the total positive predictions (True Positive and False Positive).

$$\textit{Precision} = \frac{TP}{(TP + FP)}$$

■ IV. Recall or Sensitivity

- It is also similar to the Precision metric; however, it aims to calculate the proportion of actual positive that was identified incorrectly. It can be calculated as True Positive or predictions that are actually true to the total number of positives, either correctly predicted as positive or incorrectly predicted as negative (true Positive and false negative).
- The formula for calculating Recall is given below:

$$\text{Recall} = \frac{TP}{TP + FN}$$

■ When to use Precision and Recall?

- From the above definitions of Precision and Recall, we can say that recall determines the performance of a classifier with respect to a false negative, whereas precision gives information about the performance of a classifier with respect to a false positive.
- So, if we want to minimize the false negative, then, Recall should be as near to 100%, and if we want to minimize the false positive, then precision should be close to 100% as possible.
- In simple words, *if we maximize precision, it will minimize the FP errors, and if we maximize recall, it will minimize the FN error.*

V. F-Scores

- F-score or F1 Score is a metric to evaluate a binary classification model on the basis of predictions that are made for the positive class. It is calculated with the help of Precision and Recall. It is a type of single score that represents both Precision and Recall. So, ***the F1 Score can be calculated as the harmonic mean of both precision and Recall, assigning equal weight to each of them.***
- The formula for calculating the F1 score is given below:

$$F1 - score = 2 * \frac{precision * recall}{precision + recall}$$

■ When to use F-Score?

- As F-score make use of both precision and recall, so it should be used if both of them are important for evaluation, but one (precision or recall) is slightly more important to consider than the other.
- For example, when False negatives are comparatively more important than false positives, or vice versa.

2. Performance Metrics for Regression

- The metrics used for regression are different from the classification metrics.
- It means we cannot use the Accuracy metric (explained above) to evaluate a regression model; instead, the performance of a Regression model is reported as errors in the prediction.
- Metrics that are used to evaluate the performance of Regression models.
 - **Mean Absolute Error**
 - **Mean Squared Error**
 - **R2 Score**
 - **Adjusted R2**

■ I. Mean Absolute Error (MAE)

- Mean Absolute Error or MAE is one of the simplest metrics, which measures the absolute difference between actual and predicted values, where absolute means taking a number as Positive.
- To understand MAE, let's take an example of Linear Regression, where the model draws a best fit line between dependent and independent variables. To measure the MAE or error in prediction, we need to calculate the difference between actual values and predicted values. But in order to find the absolute error for the complete dataset, we need to find the mean absolute of the complete dataset.

- The below formula is used to calculate MAE:

$$MAE = 1/N \sum |Y - Y'|$$

Here, Y is the Actual outcome, Y' is the predicted outcome, and N is the total number of data points.

II. Mean Squared Error

- Mean Squared error or MSE is one of the most suitable metrics for Regression evaluation. It measures the average of the Squared difference between predicted values and the actual value given by the model.
- Since in MSE, errors are squared, therefore it only assumes non-negative values, and it is usually positive and non-zero.
- Moreover, due to squared differences, it penalizes small errors also, and hence it leads to over-estimation of how bad the model is.
- MSE is a much-preferred metric compared to other regression metrics as it is differentiable and hence optimized better.

- The formula for calculating MSE is given below:

$$MSE = 1/N \sum (Y - Y')^2$$

Here, Y is the Actual outcome, Y' is the predicted outcome, and N is the total number of data points.

III. R Squared Score

- R squared error is also known as Coefficient of Determination, which is another popular metric used for Regression model evaluation. The R-squared metric enables us to compare our model with a constant baseline to determine the performance of the model. To select the constant baseline, we need to take the mean of the data and draw the line at the mean.
- The R squared score will always be less than or equal to 1 without concerning if the values are too large or small.

$$R^2 = 1 - \frac{MSE(Model)}{MSE(Baseline)}$$

IV. Adjusted R Squared

- Adjusted R squared, as the name suggests, is the improved version of R squared error.
- We can calculate the adjusted R squared as follows:

$$R_a^2 = 1 - \left[\left(\frac{n-1}{n-k-1} \right) \times (1 - R^2) \right]$$

Here,

n is the number of observations

k denotes the number of independent variables

and R_a^2 denotes the adjusted R^2

